Test_r

2025-10-29

```r
library(tidyverse)
```

Introduction

The chosen dataset comes from tidytuesday and is a dataset regarding Pokemons. It can be found here: https://github.com/rfordatascience/tidytuesday/blob/main/data/2025/2025-04-01

Data presentation

A summery of the data set can be seen below. From this information, it is possible to see which variables makes sense to do statistical test's on.

As an example "id" and "species_id" does not make sense to use in these test's, because it is a index value more that data value.

Therefore for this assignment, the chosen variables for further statistical investigation are the Pokémons "attack" and "hp" data.

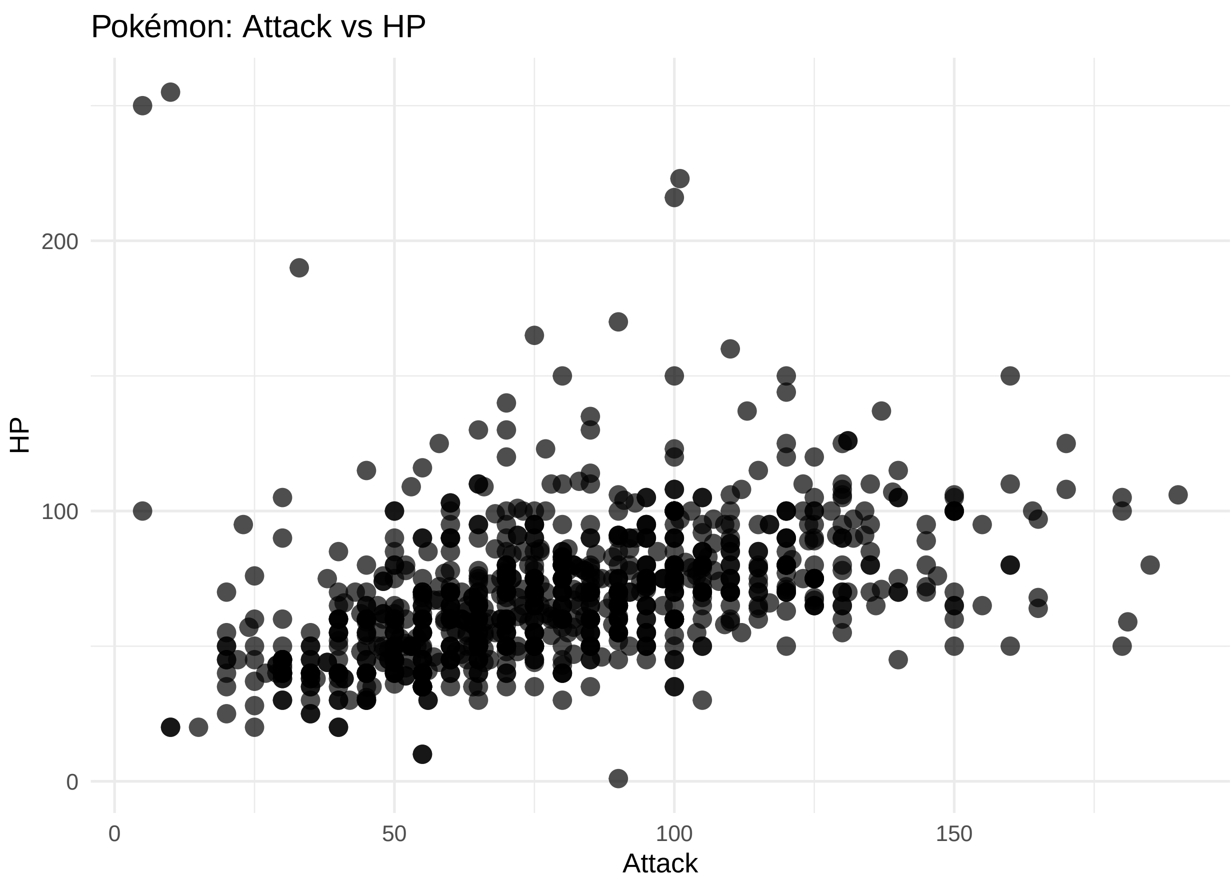
```r
summary(pokemon_df)
```

```
##        id          pokemon            species_id        height
##  Min.   :    1   Length:949         Min.   :  1.0   Min.   : 0.100
##  1st Qu.:  238   Class :character   1st Qu.:191.0   1st Qu.: 0.500
##  Median :  475   Mode  :character   Median :395.0   Median : 1.000
##  Mean   : 1900                      Mean   :400.1   Mean   : 1.228
##  3rd Qu.:  712                      3rd Qu.:615.0   3rd Qu.: 1.500
##  Max.   :10147                      Max.   :802.0   Max.   :14.500
##
##      weight       base_experience    type_1             type_2
##  Min.   :  0.10   Min.   : 36.0   Length:949         Length:949
##  1st Qu.:  8.50   1st Qu.: 68.0   Class :character   Class :character
##  Median : 28.80   Median :157.0   Mode  :character   Mode  :character
##  Mean   : 66.21   Mean   :150.5
##  3rd Qu.: 66.60   3rd Qu.:184.0
##  Max.   :999.90   Max.   :608.0
##
##        hp             attack          defense       special_attack
##  Min.   :  1.00   Min.   :  5.00   Min.   :  5.00   Min.   : 10.00
##  1st Qu.: 50.00   1st Qu.: 55.00   1st Qu.: 50.00   1st Qu.: 50.00
##  Median : 65.00   Median : 75.00   Median : 70.00   Median : 65.00
##  Mean   : 68.95   Mean   : 79.47   Mean   : 74.07   Mean   : 72.81
```

```
##  3rd Qu.: 80.00   3rd Qu.:100.00   3rd Qu.: 90.00   3rd Qu.: 95.00
##  Max.   :255.00   Max.   :190.00   Max.   :230.00   Max.   :194.00
##
##  special_defense      speed          color_1            color_2
##  Min.   : 20.00   Min.   :  5.00   Length:949         Length:949
##  1st Qu.: 50.00   1st Qu.: 45.00   Class :character   Class :character
##  Median : 70.00   Median : 65.00   Mode  :character   Mode  :character
##  Mean   : 72.22   Mean   : 69.02
##  3rd Qu.: 90.00   3rd Qu.: 90.00
##  Max.   :230.00   Max.   :180.00
##
##    color_f          egg_group_1        egg_group_2          url_icon
##  Length:949         Length:949         Length:949         Length:949
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##  generation_id    url_image
##  Min.   :1.000   Length:949
##  1st Qu.:2.000   Class :character
##  Median :4.000   Mode  :character
##  Mean   :3.695
##  3rd Qu.:5.000
##  Max.   :7.000
##  NA's   :147
```

To visualize the data, one can plot the two variables against each other. As seen below:

```r
ggplot(pokemon_df, aes(x = attack , y = hp)) +
  geom_point(alpha = 0.7, size = 3) +
  theme_minimal() +
  labs(
    title = "Pokémon: Attack vs HP",
    x = "Attack",
    y = "HP"
  )
```



We setup a test, that tests if there is a correlation between the two variables.

Here our null hypothesis is that the more HP a pokémon has the more attack it will have as well

```r
cor_result <- cor.test(pokemon_df$attack, pokemon_df$hp)

cor_result$estimate  # correlation coefficient
```

```
##       cor
## 0.4266479
```

```r
cor_result$p.value   # p-value
```

```
## [1] 2.904688e-43
```

```r
linear_model <- lm(attack ~ hp, data=pokemon_df)
summary(linear_model)
```

```
##
## Call:
## lm(formula = attack ~ hp, data = pokemon_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

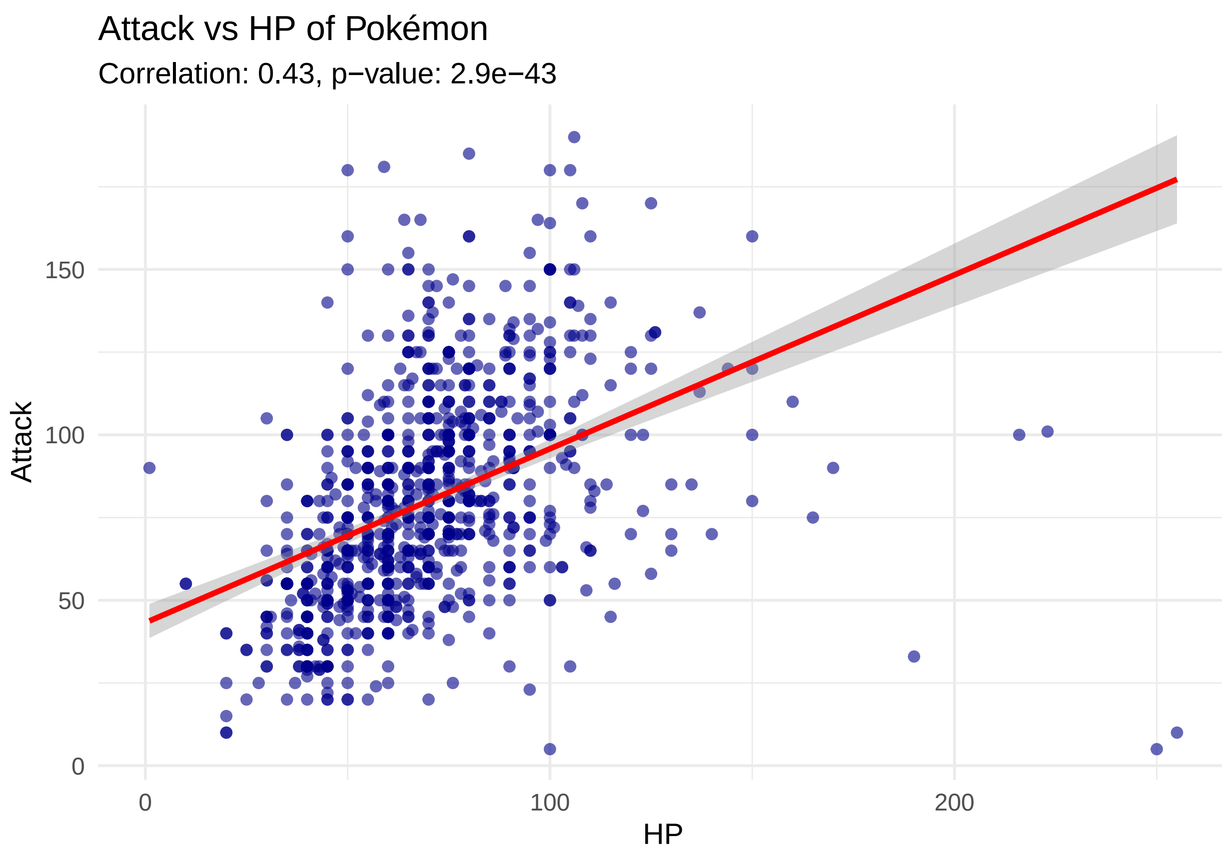
```
## -169.623  -18.357   -3.277   17.095  110.490
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  43.2319     2.6669   16.21   <2e-16 ***
## hp            0.5256     0.0362   14.52   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 28.91 on 947 degrees of freedom
## Multiple R-squared:  0.182,  Adjusted R-squared:  0.1812
## F-statistic: 210.7 on 1 and 947 DF,  p-value: < 2.2e-16
```

From this we have a very low p-value, meaning that we will have to reject our null hypothesis. Furthermore if one where to fit a linear model to this data we can see the model fit very poorly, both visually as seen below, but it is also possible to examine the correlation coefficient of the fit.

```r
library(ggplot2)

ggplot(pokemon_df, aes(x = hp, y = attack)) +
  geom_point(alpha = 0.6, color = "darkblue") +       # scatter points
  geom_smooth(method = "lm", se = TRUE, color = "red") +  # regression line
  theme_minimal() +
  labs(
    title = "Attack vs HP of Pokémon",
    subtitle = paste0("Correlation: ", round(cor_result$estimate, 2),
                      ", p-value: ", signif(cor_result$p.value, 3)),
    x = "HP",
    y = "Attack"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



To make the next test a combination of the offensive and defensive stats respectively of the pokémon have been made. Considering attack, special_attack and speed offensive traits, and defense, special_defense and hp to be defensive traits. The combination of these is made with a simple summation, and save in new columns called offensive and defensive. This is done based on the hypothesis that a pokémon with with combined higher offensive stats would have lower defensive stats and vise versa.

```
pokemon_df <- pokemon_df %>%
  mutate(offensive = attack + special_attack + speed)

pokemon_df <- pokemon_df %>%
  mutate(defensive = defense + special_defense + hp)
```

Using the linear model like before we test to see if there is a correlation between combined offensive and defensive stats of a pokémon. Again we get a p-value in the 2e-16 range which means we see no statistical correlation between the offensive and defensive stats.

```
linear_model_off_deff <- lm(offensive ~ defensive, data=pokemon_df)
summary(linear_model_off_deff)
```

```
##
## Call:
## lm(formula = offensive ~ defensive, data = pokemon_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -337.12  -41.40   -3.63   36.16  355.30
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 106.83005    7.15629   14.93   <2e-16 ***
## defensive     0.53185    0.03183   16.71   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 63.79 on 947 degrees of freedom
## Multiple R-squared:  0.2277, Adjusted R-squared:  0.2269
## F-statistic: 279.3 on 1 and 947 DF,  p-value: < 2.2e-16
```

To visualize the same code is reused.

```r
cor_result <- cor.test(pokemon_df$defensive, pokemon_df$offensive)

cor_result$estimate  # correlation coefficient
```

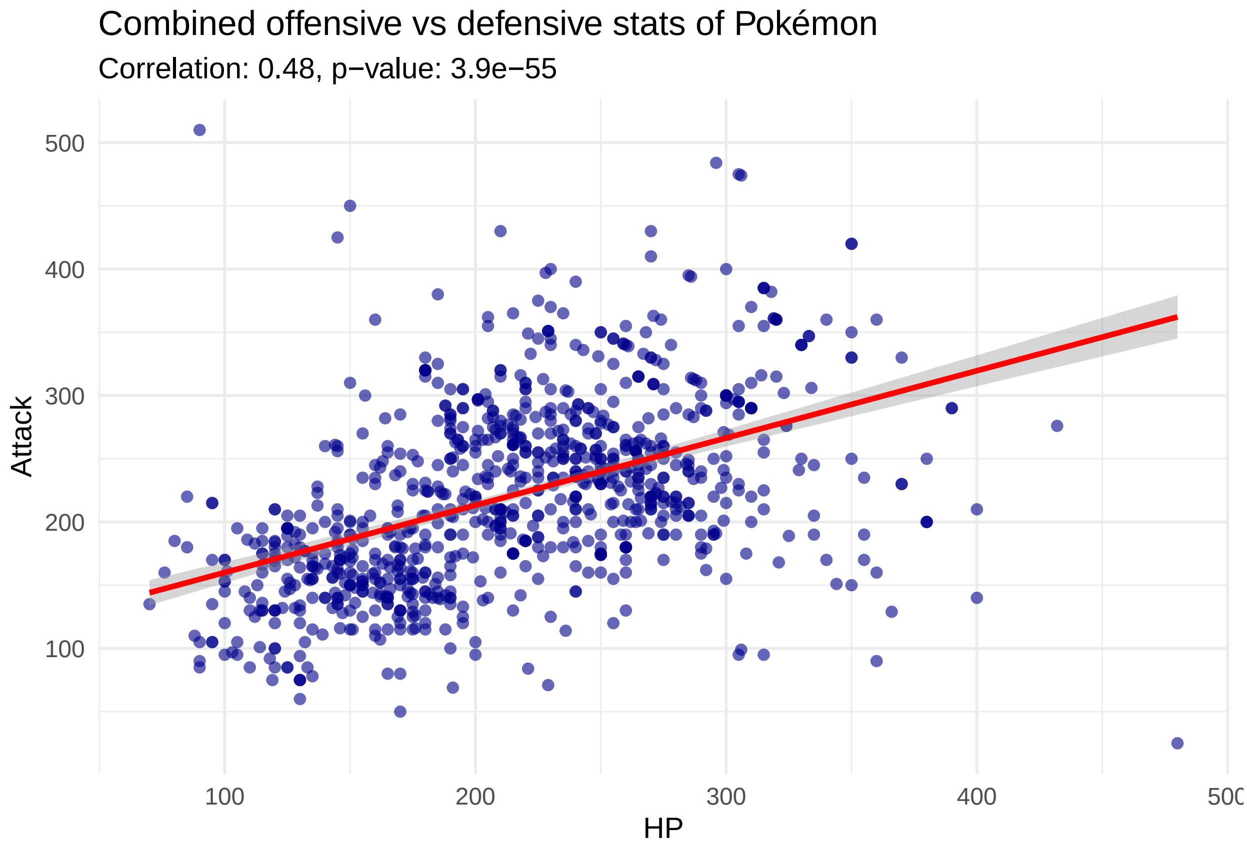
```
##      cor
## 0.477215
```

```r
cor_result$p.value   # p-value
```

```
## [1] 3.899306e-55
```

```r
ggplot(pokemon_df, aes(x = defensive, y = offensive)) +
  geom_point(alpha = 0.6, color = "darkblue") +       # scatter points
  geom_smooth(method = "lm", se = TRUE, color = "red") +  # regression line
  theme_minimal() +
  labs(
    title = "Combined offensive vs defensive stats of Pokémon",
    subtitle = paste0("Correlation: ", round(cor_result$estimate, 2),
                      ", p-value: ", signif(cor_result$p.value, 3)),
    x = "HP",
    y = "Attack"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```


Combined offensive vs defensive stats of Pokémon
Correlation: 0.48, p−value: 3.9e−55
Attack
500
400
300
200
100
HP
100
200
300
400
500

“‘